

W002 — Colour-Line Decomposition & Validation Report

Project: MINERVA QCELL P&ID — colour-line-first engineering process model

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Wave: W002 — Colour Line Decomposition & Validation · Status: ACTIVE

Pipeline: `src/abacus_svg_pid/parser.py` + `cli.py` (pure standard library)

Geometry, arrow tracing and sequential component ordering are intentionally

DEFERRED to W004 (placeholders `"DEFERRED_W004"` in the model). W003

(layer/mechanical-section engine), UI, PPT and CI workflows are **not** run.

1. SVG sources found

Real source drawings copied into `data/svg/` as the first action:

File	Status
<code>PFD-PID MINERVA QCELL-LB.svg</code>	✓ loaded
<code>PFD-PID MINERVA RFCELL seen by ACR.svg</code>	✓ loaded

SVG files loaded: 2 (STOP gate ≥ 2 satisfied).

2. PDF / PPT references found

File	Location
<code>PFD-PID_MINERVA_QCELL-LB.pdf</code>	<code>data/pdf/</code> ✓
<code>QCELL_P&ID.pptx</code>	<code>data/ppt/</code> ✓

3. Colour-extraction correctness (bug fix)

Colour/style is read with **inline-style precedence**:

`style.get(key)` or `elem.attrib.get(key)` — inline `style="stroke:..."` overrides the presentation attribute. (The previous reversed order was the defect this wave fixes.)

The “effective process colour” of each element is its stroke when present, else its fill (the QCELL drawing draws most lines as filled paths where `stroke == fill`).

4. Colour bins detected → canonical process codes

Clustering is by **RGB colour-distance to canonical anchors** (threshold 90), not exact-hex matching, so shade variants collapse onto the right line.

Canonical bin	Anchor hex	Process code(s)	Role
BLUE / NAVY	#0000ff / #000080	A / A'	4.5 K main + internal branch
CYAN / TEAL	#00ffff / #008080	B / B'	2 K internal line
GREEN	#00ff00 / #008000	W	coupler (splits from BLUE A in QM)
OLIVE	#808000	S	warm S line
GREY	#999999 / #808080	V	vent line (per module, to outside)
RED / ORANGE	#ff0000 / #ff8000	D / E	warm/cold manifold
BLACK	#000000 / #1a1a1a	structure	boundary / symbols / unknown

5. Path / line elements per canonical process code

Process code	Element count
A (4.5 K main, BLUE)	171
A' (internal branch, NAVY)	70
B (2 K internal, CYAN)	105
W (coupler, GREEN)	160
S (S line, OLIVE)	125
V (vent, GREY)	116
D (manifold, RED)	116
E (manifold, ORANGE)	9
structure / unknown (BLACK + other)	982

6. Unique stroke colours

15 unique `#rrggbb` stroke colours across the two drawings;

220 unique `(stroke_hex, stroke_width)` pairs catalogued in `data/model/colour_inventory.json`.

7. Text labels per file

File	Text nodes
PFD-PID MINERVA QCELL-LB.svg	566
PFD-PID MINERVA RFCELL seen by ACR.svg	128

8. Boundaries / mechanical sections detected (via label scan)

Boundary	Example labels matched
QM	"scope QM", "Cryomodule (QM)", "QCELL-QM"
Jumper	"Jumper"
QVB	"scope QVB", "scope QVB (AUB)", "QVB IN-VAC"
QINFRA	"scope QINFRA", "QINFRA - Implementation by NFS"
vacuum barrier	"vacuum barrier"

All five expected scope/boundary features are present in the QCELL drawing.

9. Unresolved colours

Black family (#000000 ×728, #1a1a1a ×142) is treated as **structure / boundary / symbols** (expected, not a process line).

Truly **unmapped "other"** colours (outside the canonical mapping, flagged for engineering review — they are isolated into `unknown_black_or_other.json`):

Hex	Occurrences	Note
#ff00ff (magenta)	78	Largest unresolved bin — instrument-air / annotation line; needs a canonical assignment.
#bf512e	8	Brown/terracotta — possible RED/ORANGE tint beyond threshold.
#d35f5f	8	Dusky red — possible RED tint.
#ffffff	10	White (background fill); non-process.
#80b3ff	4	Light-blue tint — possible BLUE fill.
#55ff99	3	Light-green tint — possible GREEN fill.
#800000	1	Maroon — possible RED tint.

10. Confidence notes

- **High confidence:** A, A', B, W, S, V, D mappings — exact anchor hits (distance 0).
- **Medium:** E (ORANGE) only 9 elements; verify against legend “E 60 K; 13 bar”.
- **Low / flagged:** the seven “other” colours above; magenta in particular has a material footprint (78 elements) and must be resolved before W008 reassembly.
- `arrows_detected` and `sequential_components` are **DEFERRED_W004** — line ordering is preserved as-found (no geometric re-ordering performed).
- Tag association is a **label-seed** best-effort only; validated geometric proximity association is W005.

Success criteria (runtime counts — all non-zero)

#	Criterion	Value
1	SVG files loaded	2
2	Unique stroke colours	15
3	Path elements per process code	A 171 · A' 70 · B 105 · W 160 · S 125 · V 116 · D 116 · E 9 · unknown 982
4	Text nodes per file	QCELL 566 · RFCELL 128
5	Colours that could not be mapped	7 “other” hexes (magenta #ff00ff ×78 + 6 tints); black family = structure

Generated artefacts: data/model/colour_inventory.json , data/model/line_model.json , data/model/lines/{blue_A, cyan_B_2K, green_W_coupler, grey_V_vent, olive_S_line, red_orange_D_E, unknown_black_or_other}.json , reports/navigation.json , publish/colour_line_collage.html (+ publish/assets/*.png).